

Data Warehouses and Data Mining Systems

Laboratory 2: Report content

1 Analysis

Analysis of the given problem and data set – extended description of data structures, columns, tables, file formats characterizing the source data chosen during the first laboratory.

2 Operational database model

Development of the logical model for a operational database. It should be done using all the knowledge gathered during the Fundamentals of Database Systems. Do not think about the data warehouses yet, do not divide data into facts nor dimensions – simply try to model a database that best suits the chosen datasets.

3 Data warehouse conceptual model

Development of the conceptual model for the data warehouse. This model should contain the following elements:

1. Identification of facts which are the core of the analysis and are the subject of the researched topic (e.g. sales).
2. Identification of dimensions which are the circumstances of the previously defined fact/facts (e.g. time, location etc.).
3. Classification of all (or a selected subset) columns from the original datasets into two categories:
 - a. measures – elements of facts,
 - b. dimensional attributes – elements of dimensions.
4. Drawing a conceptual model diagram which consists of facts and dimensions, surrounded by attributes.

